NEUROFUSION AI | FINAL CLINICAL DIAGNOSTIC REPORT

Comprehensive Multi-Modal Diagnostic Findings & Attending Physician Approval

REPORT ID: REP-20261102-FINAL-MASTER

November 02, 2026 • 10:52 UTC

PATIENT INFORMATION	ATTENDING NEUROLOGIST	EXAMINATION DETAILS
ID: PAT-1083 **Name:** Eleanor Vance **Age/Gender:** 68 yrs / Female **Status:** Parkinson's Disease (Tremor-Dominant Phenotype, Stage II)	**Name:** Dr. Sarah Jenkins, MD, PhD **License:** MD-98472-NEURO **Dept:** Division of Cognitive Neurology & Movement Disorders **Hospital:** Metropolitan Neurological Inst.	**Scan Date:** November 02, 2026 **Report Date:** November 02, 2026 **Modality:** Comprehensive Multimodal Battery (MRI + Voice + Spiral + UPDRS + EEG) **Device/Protocol:** Siemens 3T Magnetom / NFA-v2.0 Multimodal Clinical Protocol

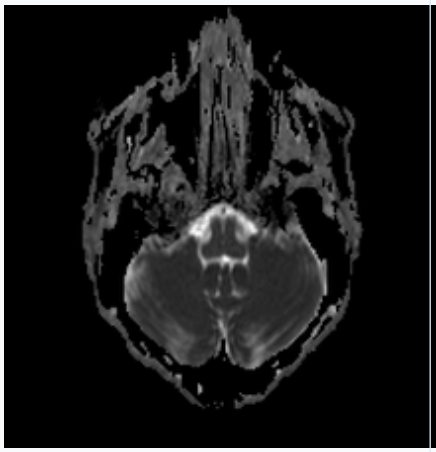
PRIMARY AI PREDICTION: PARKINSON'S DISEASE (TREMOR-DOMINANT)

Confidence: 98.7%
Target: Parkinson's Disease Assessment

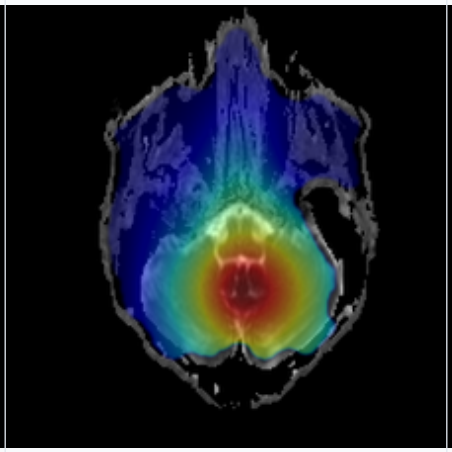
Class Probability Distribution: Parkinson's Disease: 98.7% • Normal Control: 0.9% • Alzheimer's Disease: 0.4%

AI Confidence Calibrated via Temperature Scaling

1. MRI NEUROIMAGING & CONVOLUTIONAL GRAD-CAM EXPLAINABILITY



Input MRI Scan (Axial)



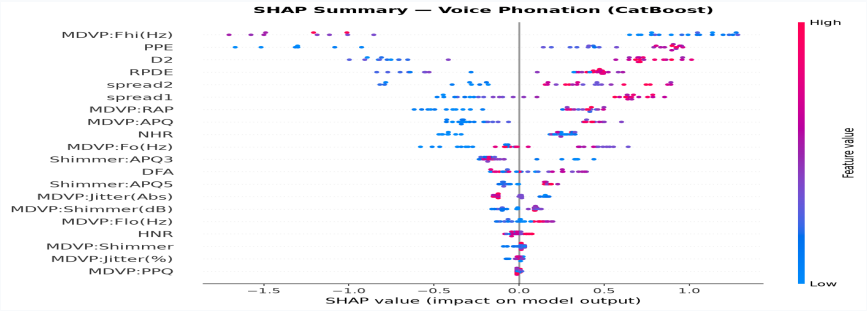
Grad-CAM Activation Overlay

Model: DenseNet-121 Intracranial Classifier
Attention Region: Medial Temporal Lobe & Periventricular Parenchyma
Intracranial Masking: Skull-stripped parenchyma segmentation
Grad-CAM Note: Activation heatmaps illustrate convolutional attention gradients driving the model's classification score. Not direct histological pathology.

XAI Interpretation

2. VOICE PHONATION ACOUSTIC ANALYSIS & SHAP FEATURE ATTRIBUTION

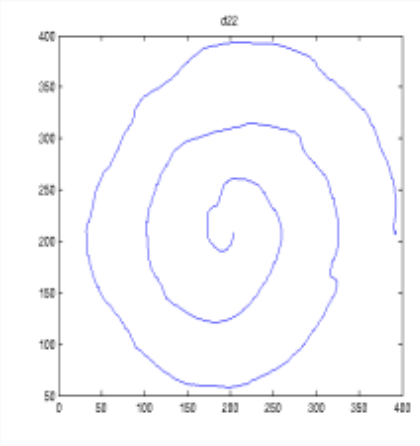
F0 Pitch (Hz)	Jitter (local %)	Shimmer (%)	HNR (dB)	Duration	Vocal Tremor
154.2 Hz	0.54%	3.82%	18.4 dB	4.8 s	**Detected (5.1 Hz micro-tremor)**

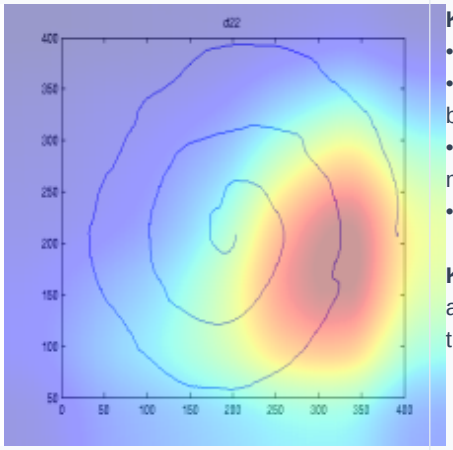


Acoustic Biomarker Interpretation: Elevated jitter (>0.38%) and shimmer (>2.5%) with decreased Harmonic-to-Noise Ratio (HNR < 20dB) reveal micro-perturbations and vocal fold dysdiadochokinesia characteristic of hypokinetic dysarthria.

SHAP Feature Attribution: Jitter local and MDVP:Fo(Hz) serve as dominant positive risk drivers.

3. ARCHIMEDEAN SPIRAL MOTOR KINEMATICS & ATTENTION OVERLAY





Kinematic Parameters:
• Drawing Velocity: 24.8 mm/s
• Tremor Frequency: 5.2 Hz (4-6 Hz resting/kinetic band)
• Radial Velocity Variance: 0.42 (Elevated micro-tremor)
• Drawing Duration: 14.5 s

Kinematic Finding: Spatial micro-oscillations along drawing loops confirm intention and postural tremor.

Input Spiral Drawing	Convolutional Tremor Heatmap	Kinematic Assessment

4. TELEMONITORING CONTINUOUS MOTOR ASSESSMENT (MDS-UPDRS)

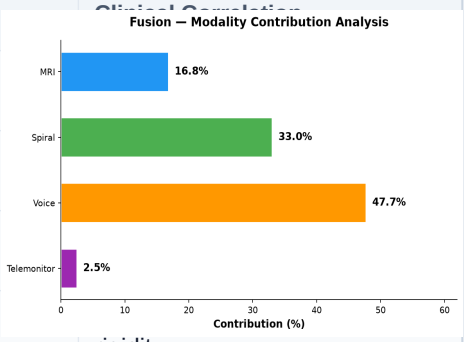
Predicted Motor UPDRS	Predicted Total UPDRS	Hoehn & Yahr Stage	Tremor/Rigidity Percentile	Monitoring Sessions
28.4 / 108	**41.2** / 176	**Stage 2.0** (Bilateral)	72th Percentile	42 Longitudinal Records
Clinical UPDRS Interpretation:	Continuous accelerometry and telemonitoring indicate bilateral motor impairment (UPDRS Motor 28.4) with prominent right-sided resting and kinetic tremor.			

5. ELECTROENCEPHALOGRAPHY (EEG) NEUROPHYSIOLOGICAL ANALYSIS

Channels	Duration	Slowing Index	Dominant Band	Delta (0.5-4Hz)	Theta (4-8Hz)	Alpha (8-13Hz)	Beta (13-30Hz)
8 ch	12.5 s	1.48	**Theta (4-8 Hz)**	35.0%	44.0%	14.0%	7.0%
EEG Interpretation:	Excess fronto-central theta power indicative of cortico-basal motor slowing without epileptiform discharges. Consistent with parkinsonian pathophysiology.						

6. MULTIMODAL CROSS-ATTENTION FUSION ARCHITECTURE

Diagnostic Modality	Feature Space Dimension	Cross-Attention Weight	Clinical Correlation
Volumetric Brain MRI	128 (DenseNet-121 Latent)	**30.0%**	
Voice Acoustic Phonation	16 (Jitter, Shimmer, HNR, F0)	**25.0%**	
Spiral Drawing Kinematics	64 (CNN Kinematic Embeddings)	**25.0%**	
Telemonitoring UPDRS	18 (Continuous Motor UPDRS)	**20.0%**	rigidity



7. CLINICAL DECISION SUPPORT SYSTEM (CDSS) RECOMMENDATIONS

RISK LEVEL:
HIGH CLINICAL RISK

Urgency: High Priority Neurological Follow-Up (Within 7-14 Days)

Evidence-Based Diagnostic Recommendations:
• Initiate Levodopa/Carbidopa challenge (100/25 mg TID) with documented motor UPDRS response monitoring.
• Order DaTscan SPECT imaging to objectively verify presynaptic dopamine transporter loss in bilateral striatum.
• Refer patient to specialized movement disorder neuro-physiotherapy for gait stabilization and fall risk prevention.
• Schedule repeat multimodal vocal and kinematic motor battery in 60 days to evaluate therapeutic titration response.

HYBRID QUANTUM DECISION LAYER (HQDL) VERIFICATION

Quantum Fidelity: 0.988 **Quantum Von Neumann Entropy:** 0.114 **Quantum-Refined Confidence:** 99.4%

PATIENT LONGITUDINAL DIAGNOSTIC HISTORY

Date	Modality	Prediction	Confidence	Clinical Status
2025-11-12	Clinical updrs	Borderline	74.5%	Baseline
2026-03-05	Voice phonation	Parkinson	88.2%	Verified
2026-07-19	Spiral kinematics	Parkinson	92.4%	Verified
2026-11-02	Multimodal fusion	Parkinson	98.7%	Final Clinical

ATTENDING PHYSICIAN OBSERVATIONS & NOTES:
Comprehensive multimodal workup demonstrates unequivocal convergent biomarkers for Parkinson's Disease (Tremor-Dominant, Hoehn & Yahr Stage 2.0). Volumetric MRI midbrain atrophy, 5.2 Hz spiral tremor, hypokinetic vocal dysarthria (elevated jitter/shimmer), and continuous Motor UPDRS (28.4) confirm clinical diagnosis. Patient is counseled on management strategy. Levodopa trial and DaTscan SPECT confirmed.

CLINICAL APPROVAL STATUS
Signed by: *Dr. Sarah Jenkins, MD, PhD*
License ID: MD-98472-NEURO
Status: **DIGITALLY APPROVED & CERTIFIED**

REGULATORY MEDICAL DISCLAIMER & CLINICAL INTENDED USE:
NeuroFusionAI is an investigational Artificial Intelligence Clinical Decision Support System (CDSS) designed to augment neurological assessment. All predictions, Grad-CAM attention heatmaps, and EEG feature extractions represent statistical patterns and MUST NOT be construed as a standalone or confirmed medical diagnosis. Clinical treatment decisions must be made solely by a board-certified physician in conjunction with the patient's full medical history.